

SET 1

Time: 30 Minutes Total Marks: 50

STUDENT DETAILS (BLOCK LETTERS)

Name : _____

Email : _____

Mobile: _____

SECTION A – PRACTICAL CODING (30 Marks)

Create ONE HTML file using on-page `<style>` and `<script>` only.

Include: Input field, Button, Output section.

Each step must use previous step result (chain logic).

TASK REQUIREMENTS:

1. Take numeric input N (10–999 only). Show error if invalid.
2. Reverse N and compute $SUM = N + \text{Reverse}$.
3. Generate mirrored number pyramid (height = digits in SUM).
4. Generate hollow diamond (height = $(SUM \% 5) + 5$).
5. If SUM is EVEN → Display EVEN STRUCTURE.
If SUM is ODD → Display ODD STRUCTURE.

SAMPLE ILLUSTRATION

INPUT

N = 123

PROCESS

Reverse = 321

SUM = 123 + 321 = 444 (EVEN)

Digits in SUM = 3 → Pyramid Height = 3

```
  1
 121
12321
```

Diamond Height = $(444 \% 5) + 5 = 9$

```
    *
   * *
  *   *
 *     *
*       *
 *     *
  *   *
   * *
    *
```

OUTPUT

EVEN STRUCTURE

SECTION B – MCQ (20 Marks)

Q1. LIFO principle is used in:

- A) Queue
- B) Stack ✓
- C) Graph
- D) Tree

Q2. Primary Key ensures:

- A) Table relation
- B) Unique row identity ✓
- C) Encryption
- D) Fast delete

Q3. Database index improves:

- A) Insert speed
- B) Retrieval speed ✓
- C) Storage
- D) Security

Q4. Stack overflow occurs when:

- A) Removing empty
- B) Adding to full ✓
- C) Looping
- D) Timeout

Q5. INNER JOIN returns:

- A) Left rows
- B) Right rows
- C) Matching rows ✓
- D) All rows

Q6. HTTP 404 indicates:

- A) Success
- B) Not Found ✓
- C) Forbidden
- D) Crash

Q7. DNS converts:

- A) Domain to IP ✓
- B) IP to MAC
- C) URL to HTML
- D) HTTP to TCP

Q8. Git command to send commits:

- A) git pull
- B) git push ✓
- C) git fetch
- D) git clone

Q9. Thread is:

- A) Program
- B) Execution unit ✓
- C) Server
- D) Query

Q10. Method overriding represents:

- A) Encapsulation
- B) Abstraction
- C) Inheritance
- D) Polymorphism ✓